

SplitSmart

What it will do:

SplitSmart helps users divide shared expenses among a group. The system calculates individual expenses, allows users to create custom payment plans, and maintains payment records to ensure organized financial tracking.

Who is the audience:

The application serves college students who live together and friends who share costs for common expenses such as dining out, traveling, and paying bills.

How it will be used:

Users create a group, add participants, and enter expenses (including who paid and how much). The application computes individual shares for all users while showing their account balances. Users can update the app as payments are made.

An example of how it will work:

A group of friends goes out to eat. One person pays \$60 for dinner, and another person pays \$20 for drinks. They enter both expenses into SplitSmart and add all group members. The app calculates how much each person needs to pay, and it shows who has debt and who should receive payment. People mark their completed payments to friends after they receive repayment, which helps them maintain accurate records.

Key Requirements Covered

- Object-Oriented: Classes such as User, Expense, and Group
- User Input: Enter names, amounts, and payments
- GUI Output: Displays balances and summaries visually
- Decision Making: Determines how to split costs (even vs uneven)
- Iteration: Loops through users and expenses to calculate totals
- Arrays: Stores lists of users and expenses